

Module 7: Deploying and Managing Software Updates with Overview & Prerequisites.

1. Overview of Software Updates in SCCM

- **Software Updates** in SCCM are used to keep all devices **patched and secure**.
- SCCM integrates with **Microsoft Update server** to download update metadata and deploy updates.
- Admins can control **which updates** → **go to which devices** → **at what time**.

2. Prerequisites for Software Updates

- 1. Windows Server Update Services (WSUS)**
 - Must be installed and integrated with SCCM.
- 2. Software Update Point (SUP) Role**
 - Installed on a site server → connects SCCM with WSUS.
- 3. Firewall Ports**
 - Allow traffic (HTTP/HTTPS) for update sync and deployment.
- 4. Internet Access (or WSUS Offline)**
 - Required for downloading updates from Microsoft Update.
- 5. Permissions**
 - SCCM server needs rights to sync and deploy updates.

◇ Example

👉 IT Admin wants to deploy **Windows 10 Security Patch (KB Update)**.

- SUP syncs with Microsoft Update.
- Update metadata downloaded.
- Patch deployed to *All Windows 10 Devices* collection.



3. Preparing SCCM Site for Updates

Before you can deploy updates, some **site preparation steps** are required:

♦ Steps:

1. **Install WSUS (Windows Server Update Services)**
 - Required for update synchronization.
2. **Add Software Update Point (SUP) Role**
 - Connects SCCM to WSUS and Microsoft Update.

3. Configure Firewall & Network Access

- Allow **HTTP/HTTPS (80/443)** and WSUS communication ports.

4. Sync with Microsoft Update

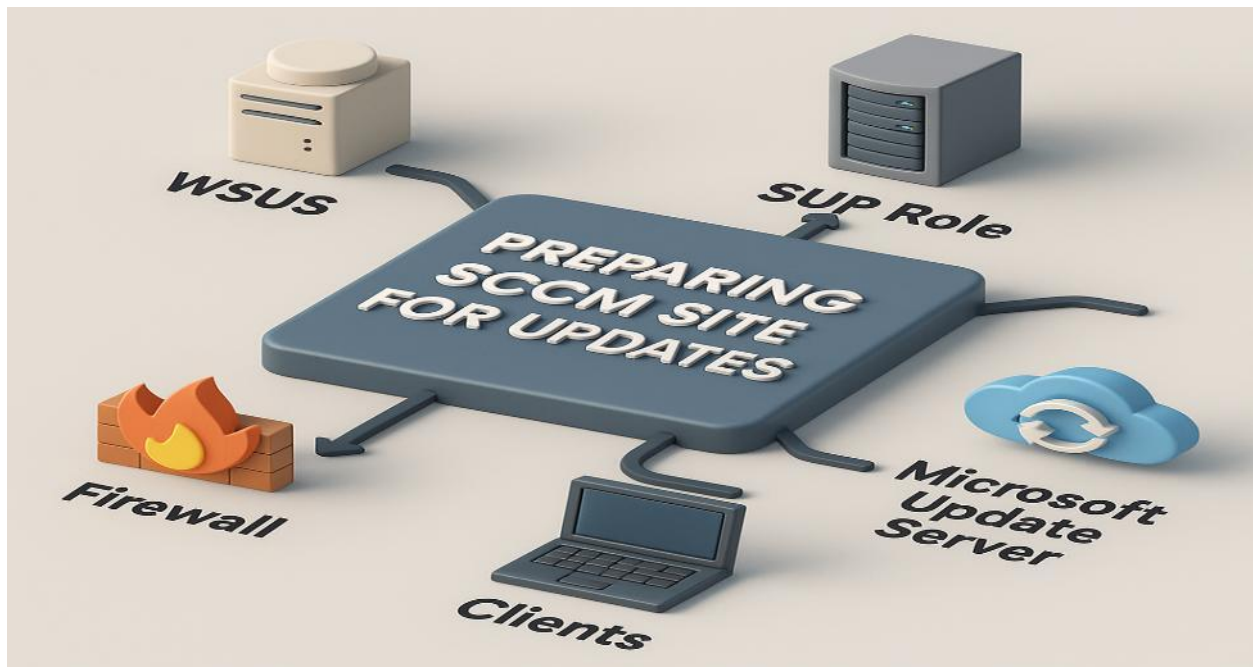
- Download update catalog/metadata.

5. Set Client Policies

- Ensure clients are configured to receive updates from SCCM.

◇ Example:

👉 Admin prepares SCCM → Adds **SUP role** → Syncs with Microsoft Update → Now SCCM can deploy **Windows 10 Security Patches**.



4. Configure Update Classifications in SCCM

Update Classifications = Categories of updates that SCCM can synchronize and deploy.

This helps admins decide **which types of updates** should be downloaded and pushed to clients.

◇ **Common Update Classifications:**

1. **Critical Updates** → Fix specific critical issues.
2. **Security Updates** → Patches for vulnerabilities.
3. **Definition Updates** → For Windows Defender / Antivirus signatures.
4. **Service Packs** → Big updates that improve stability.
5. **Feature Packs** → Add new features to Windows.
6. **Updates** → General fixes, non-critical.
7. **Update Rollups** → Collection of fixes bundled together.

◇ **Example:**

👉 Configure SCCM SUP to sync:

- **Security Updates** (for patching vulnerabilities)
- **Definition Updates** (for Defender)
- Ignore **Feature Packs** if not needed.



5. Sync with Microsoft Update Server in SCCM

◇ Purpose

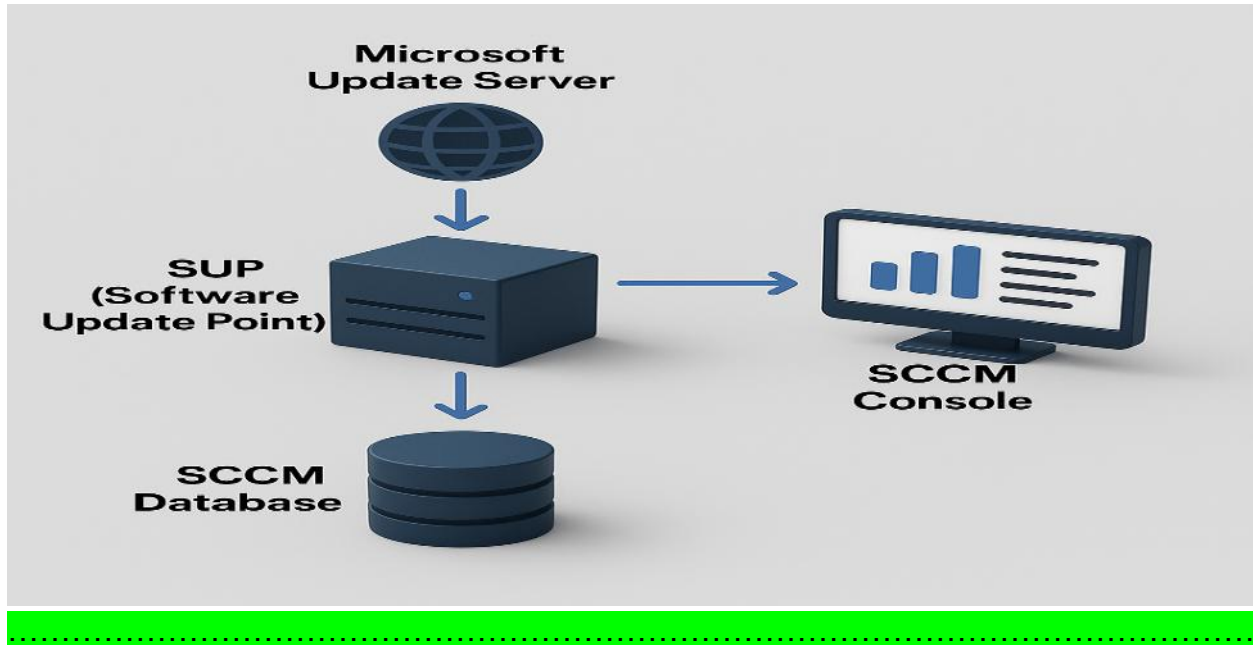
- To get the **latest update metadata** (list of available updates) from Microsoft.
- SCCM doesn't download full updates immediately → only metadata (title, description, applicability).

◇ Steps:

1. **Install & Configure SUP (Software Update Point).**
2. **Set Sync Schedule** → Daily/Weekly.
3. **SCCM SUP ↔ Microsoft Update Server** → Synchronization happens.
4. **Metadata stored in SCCM Database** → Visible in SCCM Console.

◇ Example

👉 Admin runs sync → SCCM downloads list of new patches → e.g., *Windows 10 Security Update (KB5021234)* shows in console.



6. Create Update Groups & Deployment Packages in SCCM

◇ Update Groups

- **Update Group** = A logical collection of related updates.
- Helps admins manage and deploy updates easily.
- Example:
 - *March 2025 Security Updates* group
 - *Windows 11 Feature Updates* group

◇ Deployment Packages

- **Deployment Package** = Physical container that stores update files (.cab, .msu) on a **Distribution Point (DP)**.
- Clients download updates from these DPs.
- Example:
 - Package = *March 2025 Updates Package*
 - Stored on **DP-Delhi** and **DP-Lucknow**

◇ Workflow Example

1. Select updates in SCCM Console.
2. Add them to an **Update Group**.
3. Create a **Deployment Package** → specify DP location.
4. Deploy to **Collections** (e.g., All Windows 11 PCs).



7. Deploy & Manage Updates in SCCM

◇ Deployment Process

1. **Select Update Group** → Choose updates to deploy.
2. **Target Collection** → Devices/users (e.g., *All Windows 11 PCs*).
3. **Schedule Deployment** → Immediate or at a specific time.
4. **Clients Download & Install** → From nearest DP.

◇ Management Process

- **Monitor Status** → In **Monitoring Workspace** (Success, In Progress, Failed).
- **Re-deploy Failed Updates** → Retry for clients that didn't install.
- **Supersedence / Expired Updates** → Old patches replaced or removed automatically.

◇ Example

👉 Admin deploys **March 2025 Security Updates** → to *Windows 11 PCs* → Monitors progress → Failed installs are retried.



8. Automatic Deployment Rules (ADR) in SCCM

◇ What is ADR?

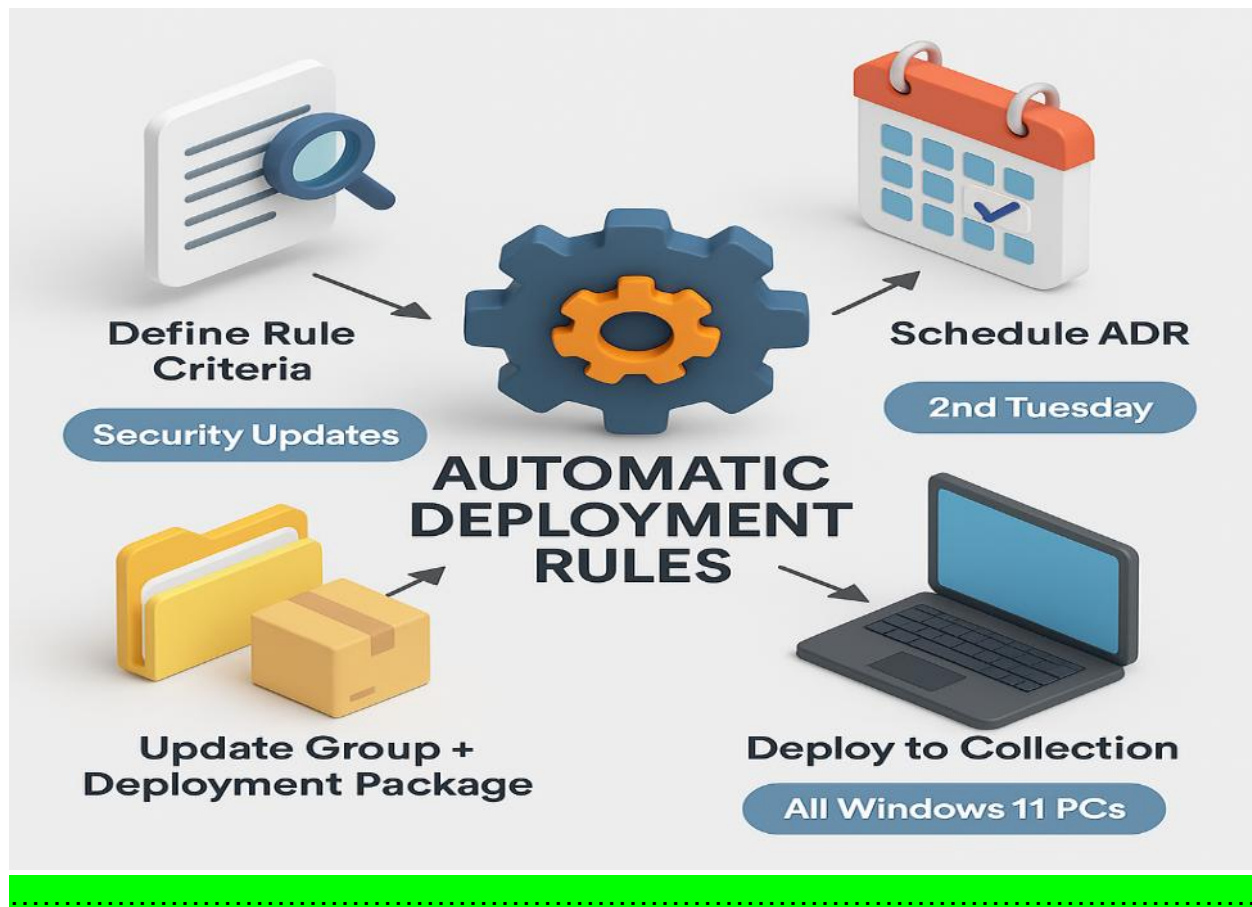
- **ADR** = Automates the process of deploying updates.
- Saves admin time → no need to manually select updates every month.

◇ How ADR Works:

- 1. Define Rule Criteria**
 - Example: Select *Security Updates* for *Windows 11*.
- 2. Schedule ADR**
 - Example: Run on **2nd Tuesday (Patch Tuesday)** every month.
- 3. SCCM Creates Update Group + Deployment Package**
 - Automatically downloads new updates.
- 4. Deploy to Collection**
 - Target e.g., *All Windows 11 PCs*.

◇ Example

👉 Create ADR → “Windows 11 Security Updates” → Runs monthly → Automatically deploys patches to all Windows 11 laptops.



9. Reports & Logs in SCCM (for Software Updates)

◇ Reports

- Found in **SCCM Console** → **Monitoring** → **Reporting** → **Reports** → **Software Updates**

- Useful

Reports:

1. **Compliance Report** → Shows which devices are updated vs missing patches.
2. **Update Status Report** → Success, Failed, or In Progress.
3. **Deployment Summary** → Overview of update deployment.

◇ Logs (Troubleshooting)

On Site Server:

- **wsyncmgr.log** → Update synchronization log.
- **wcm.log** → SUP configuration log.
- **dismgr.log** → Content distribution log.

On Client:

- **WUAHandler.log** → Windows Update agent activity.
- **UpdatesDeployment.log** → Deployment status on client.
- **UpdatesHandler.log** → Handles installation of updates.

◇ Example

👉 Admin checks **Compliance Report** → sees 90% devices patched, 10% missing updates → reviews **WUAHandler.log** on failed client.



Module 8: Managing Operating System Deployment (OSD)

1. Overview

- **OSD (Operating System Deployment)** in SCCM is used to **deploy Windows images** (e.g., Windows 10/11) to new or existing devices.
- Saves time vs manual installation.
- Useful for:
 - New PC setup
 - Upgrading OS (e.g., Windows 10 → Windows 11)
 - Re-imaging machines

2. Prerequisites

1. **Windows Deployment Services (WDS)**
 - Required for PXE boot and image deployment.
2. **Boot Images**
 - WinPE (Pre-installation environment) boot images added to SCCM.
3. **Operating System Images**
 - Captured or imported OS .wim files (e.g., Windows 11 image).
4. **Distribution Point (DP)**
 - Boot images + OS images must be distributed to DPs.

5. Task

Sequences

- Automate installation steps (partition disk, apply OS, install drivers, join domain, install SCCM client, etc.).

6. Drivers

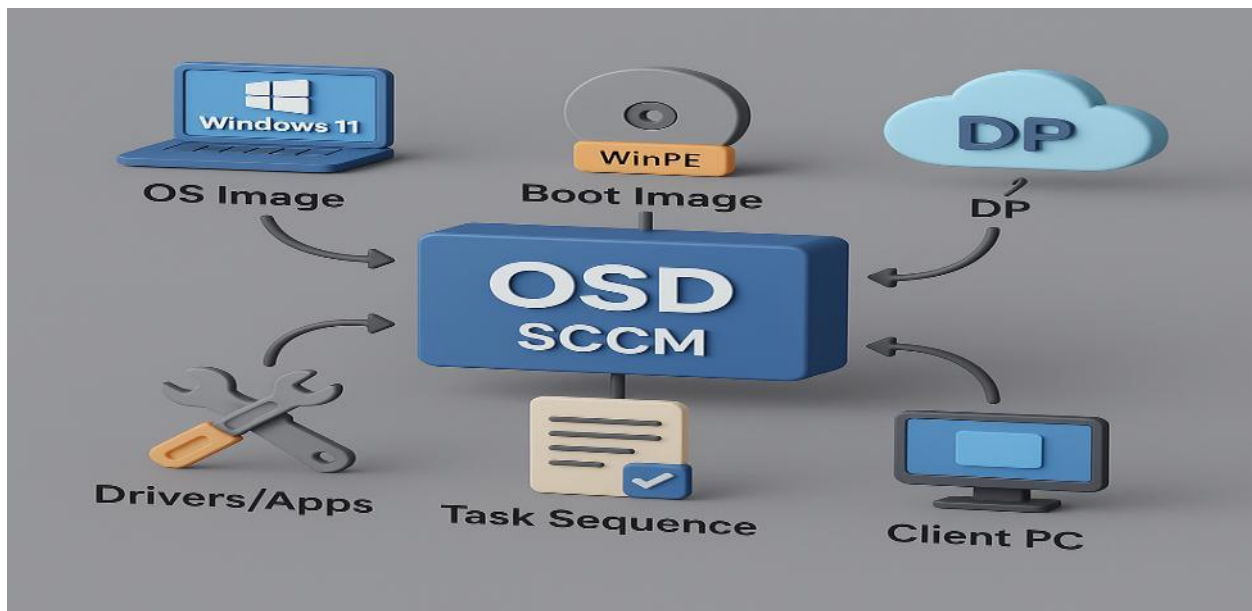
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Packages

- Device drivers added for smooth deployment.

Example

👉 IT creates a **Windows 11 Task Sequence** → Deploys to *All HR Laptops* → Machines boot via PXE → OS installs automatically with drivers + Office 365.



3. Configuring DHCP Server for OSD in SCCM

When deploying OS via **PXE boot**, DHCP must be configured properly so client machines can boot from the network.

♦ Steps to Configure DHCP

1. **Install DHCP Role** on Windows Server.

2. Create **Scope**

- Define IP range for client PCs (e.g., 192.168.1.50 – 192.168.1.200).
- Set subnet mask, default gateway, DNS.

3. PXE **Boot** **Options**

- Option **66** → **Boot Server Host Name (SCCM/WDS Server IP)**
- Option **67** → **Boot File Name (e.g., smsboot\x64\wdsnbp.com)**

4. Authorize **DHCP** **Server**

- In Active Directory → Authorize the DHCP server.

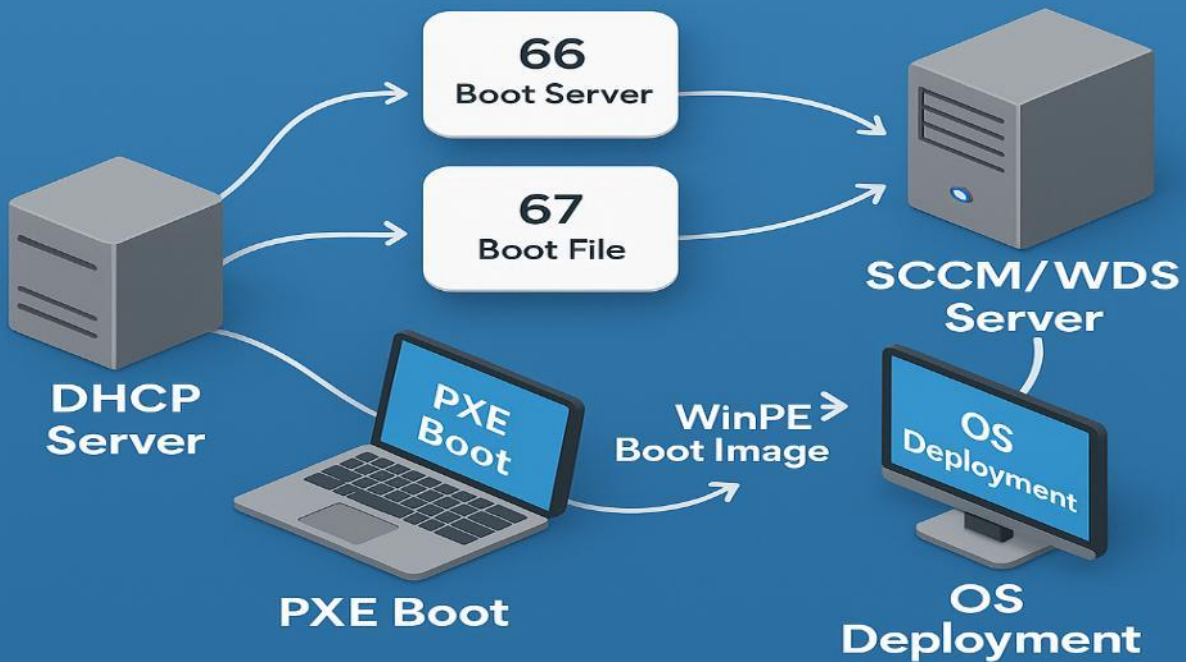
5. Test **PXE** **Boot**

- Client PC set to boot from LAN.
- Should get IP from DHCP → Contact WDS/SCCM for boot image.

◇ **Example**

👉 DHCP gives IP to a new laptop → PXE Boot → SCCM OSD starts → Windows 11 installs.

Configuring DHCP Server for OSD



4. Preparing SCCM Site for OSD (Operating System Deployment)

Before you can deploy an OS, SCCM site must be prepared with several configurations:

◆ Steps to Prepare Site

1. **Install WDS (Windows Deployment Services)**
 - Required for PXE boot support.
2. **Enable PXE on Distribution Point**
 - DP role must have PXE enabled to allow network boot.
3. **Add Boot Images**

- Import Windows PE boot images (x64, x86).
- Distribute boot images to DP.

4. **Add Operating System Images**

- Import `.wim` files (captured/customized OS image).
- Distribute them to DP.

5. **Add Drivers & Packages**

- Import hardware drivers (LAN, Storage, Chipset).
- Create driver packages and distribute to DP.

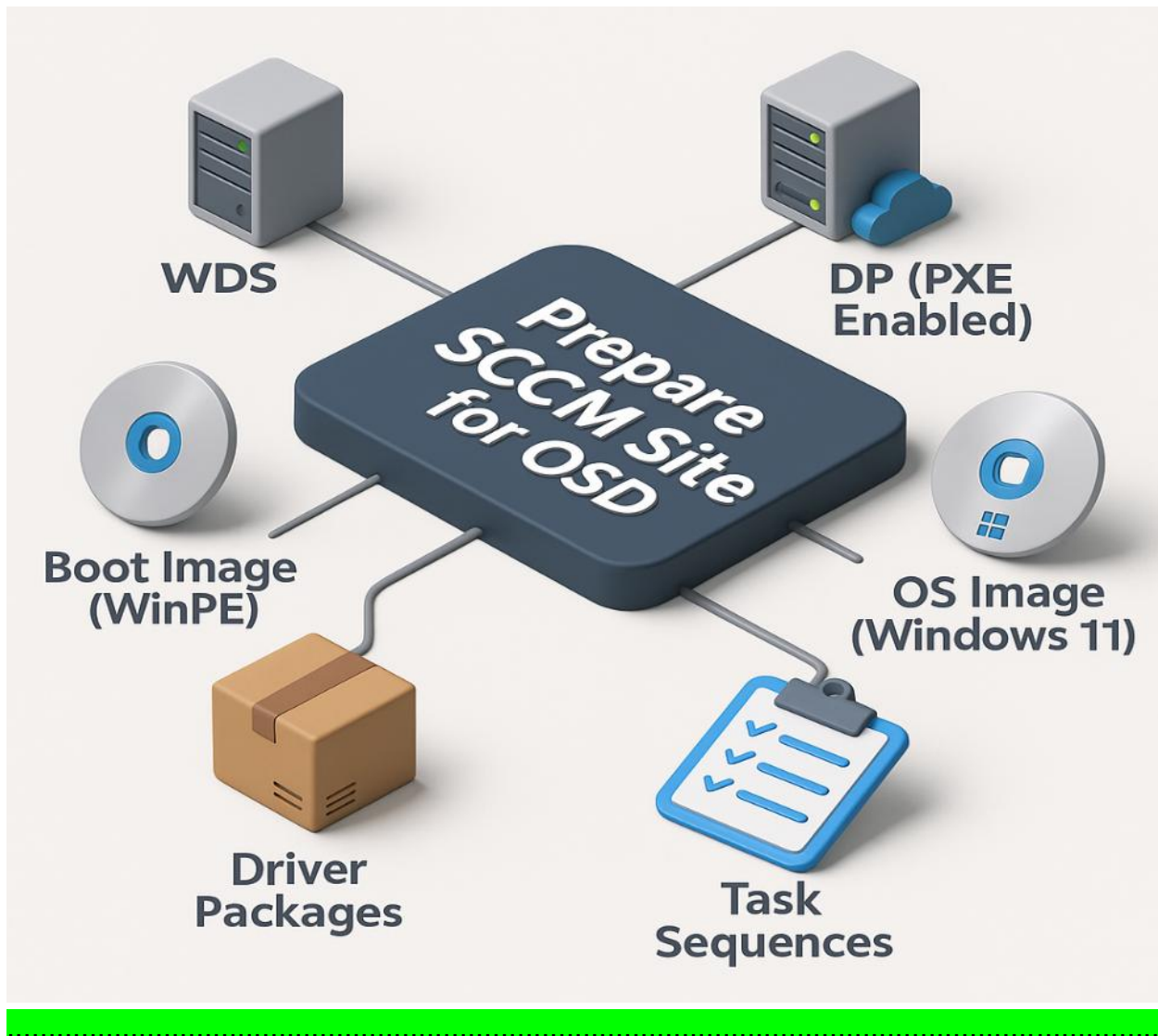
6. **Create Task Sequences**

- Automate the OSD process (partition disk → apply OS → install drivers → join domain → install SCCM client).

◇ **Example**

👉 IT prepares SCCM site with:

- Boot image = WinPE x64
- OS Image = Windows 11 Enterprise
- Task Sequence = Auto-install OS + Office 365
- DP-PXE enabled for network boot



5. Capture Reference Image in SCCM OSD

A **reference image** = A clean Windows OS image (with updates + apps) captured for deployment.

◇ Steps to Capture Reference Image

1. Prepare	Reference	Machine
○ Install	clean OS (e.g.,	Windows 11).

- Apply Windows Updates.
- Install required applications (e.g., Office, Chrome).
- No device-specific drivers.

2. Run **Capture Media**

- Create **Capture Media** in SCCM (bootable ISO/USB).
- Run on reference machine → Starts **Sysprep + Capture Wizard**.

3. Sysprep **Process**

- Generalizes the machine (removes unique identifiers).

4. **Capture Image (.wim)**

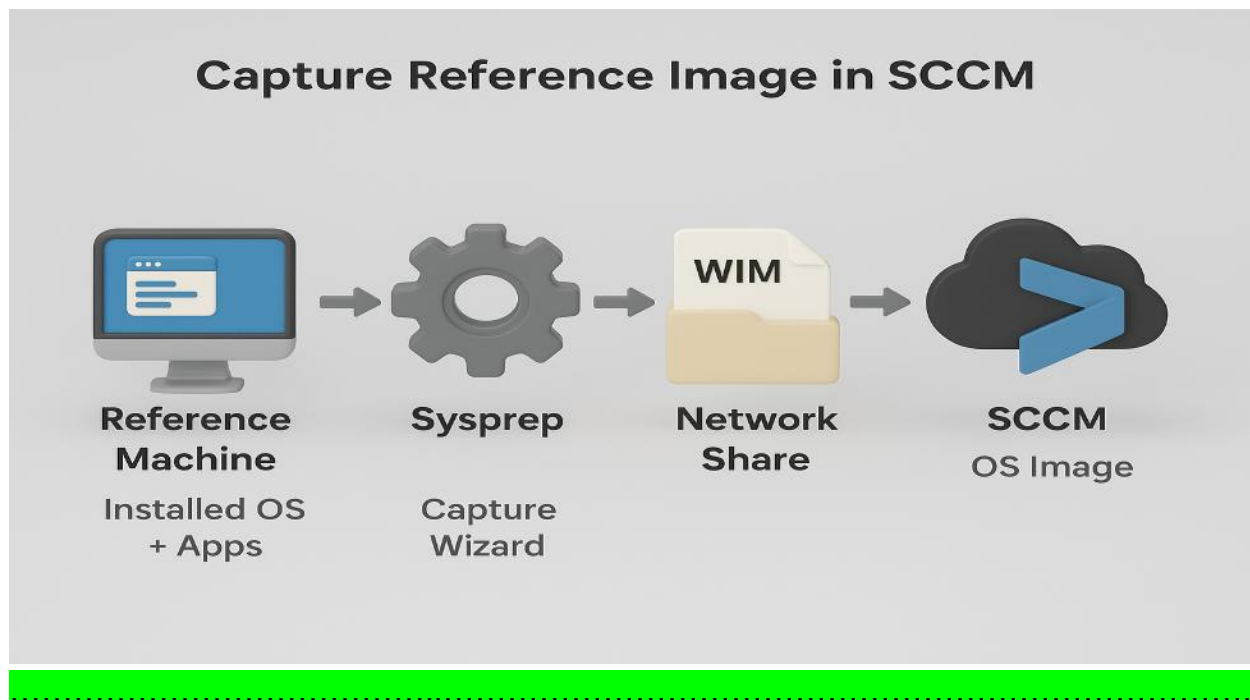
- Wizard captures the OS as a **.wim** file.
- Save to network share (e.g., `\\SCCMServer\OSImages\Win11.wim`).

5. **Import into SCCM**

- Add **.wim** file as **Operating System Image** in SCCM.
- Distribute to DPs for deployment.

◇ **Example**

👉 IT sets up a Windows 11 machine with Office & Chrome → Runs Capture Media → Gets **Win11_Ref.wim** → Imports into SCCM → Deploys to all HR laptops.



6. Create Task Sequence in SCCM (for OSD)

A **Task Sequence** = Step-by-step automated workflow that installs OS, drivers, apps, and configures settings on client PCs.

◇ Steps to Create Task Sequence

1. In **SCCM Console** → Go to:
 Software Library → Operating Systems → Task Sequences → Create Task Sequence

2. **Choose Template**

- Install an existing image package
- Build and capture a reference image

3. **Configure Steps**

- Partition disk / Format drive
- Apply Operating System Image

- Apply Drivers
- Install Applications (e.g., Office, Chrome)
- Join Domain
- Install SCCM Client
- Apply Windows Updates

4. Distribute to DP

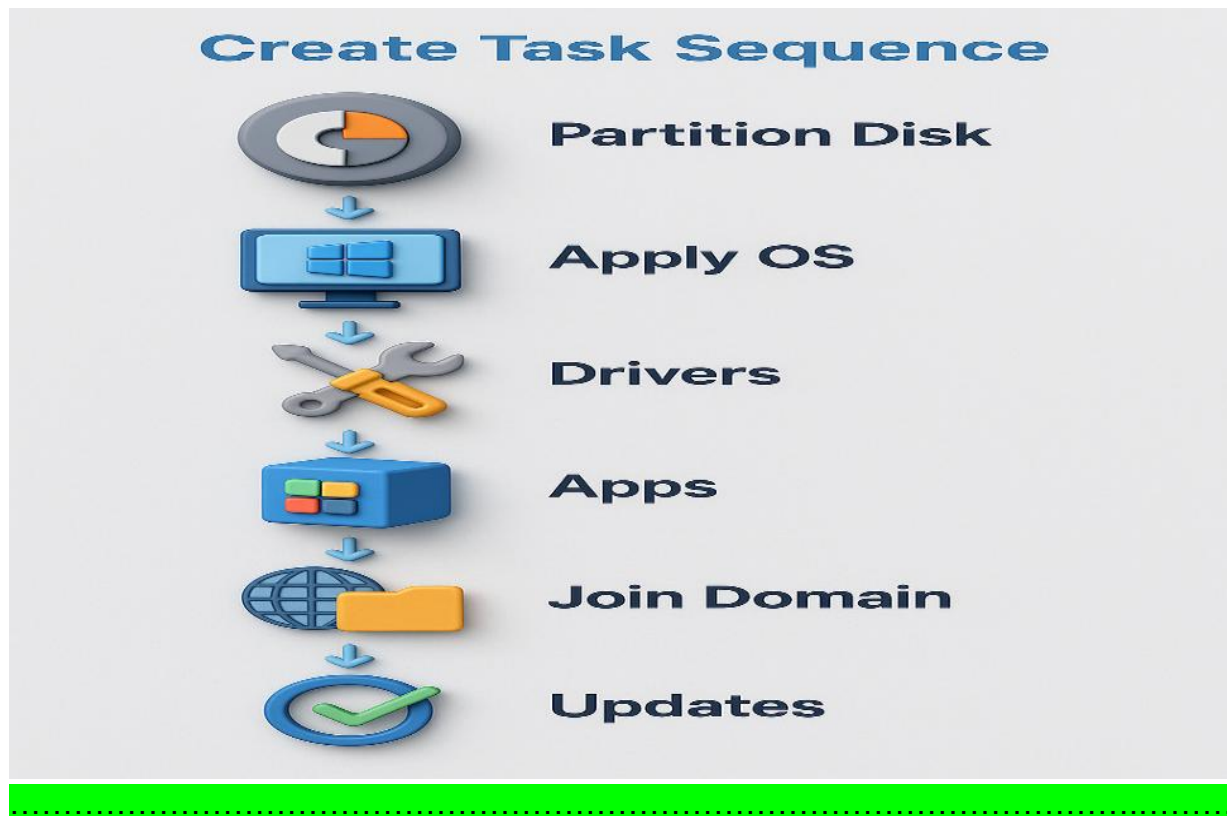
- Ensure boot image + OS image + drivers are available on DP.

5. Deploy Task Sequence

- Deploy to a collection (e.g., *All New PCs*).
- PXE boot or boot media will start the Task Sequence.

◇ Example

👉 Create a **Windows 11 Deployment Task Sequence** → Installs OS + Office 365 + Chrome + Joins Domain → All new HR laptops auto-configured.



7. Deploy OS using SCCM OSD

Once the **Task Sequence** is created, you can **deploy the OS** to target machines.

♦ Steps to Deploy OS

1. **Select** **Task** **Sequence**
 - Example: *Windows 11 Deployment Task Sequence.*
2. **Deploy** **to** **Collection**
 - Choose target devices (e.g., *All New PCs, HR Laptops*).
3. **Choose** **Deployment** **Type**
 - Available (user starts it manually from Software Center).
 - Required (auto-runs when machine boots PXE/media).

4. PXE Boot / Boot Media

- Client PC boots from network (PXE) or USB.
- Downloads **WinPE boot image** from SCCM DP.

5. Task	Sequence	Runs
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- Partition
 - Apply
 - Install
 - Join
 - Install
 - Apply
- Disk
- OS
- Image
- Drivers
- Domain
- Applications
- Updates

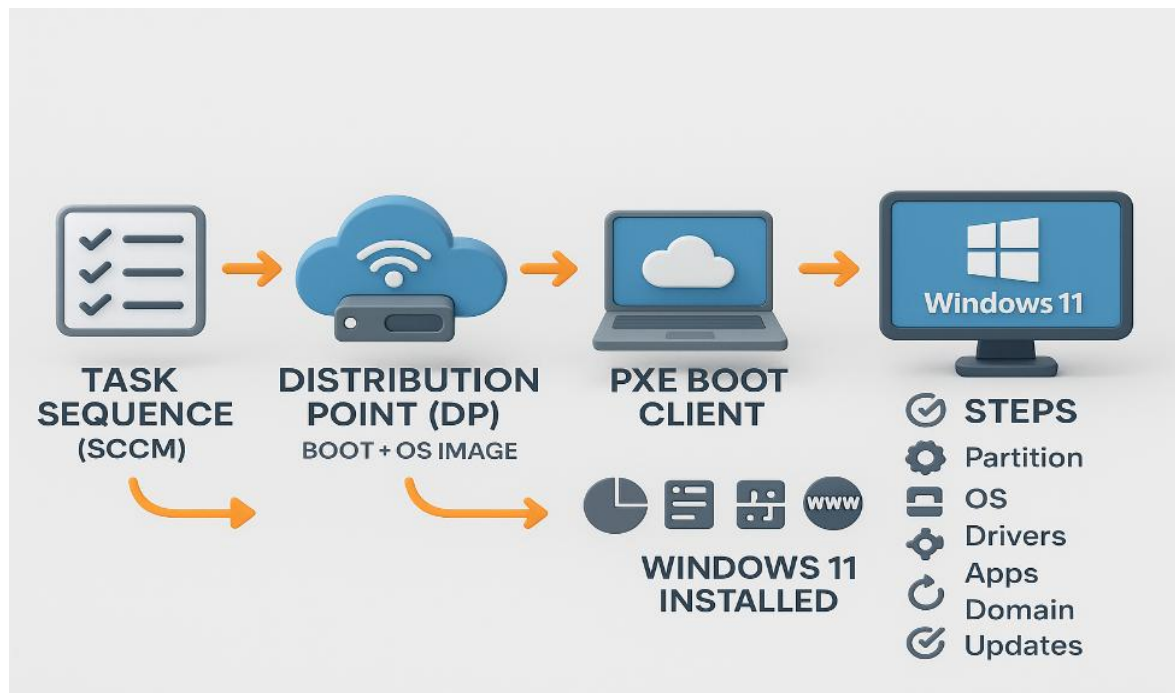
6. Deployment **Complete**

- User gets fully installed and configured OS with apps.

◇ Example

👉 IT deploys **Windows 11 OSD Task Sequence** to *HR Laptops Collection*.

- Laptops boot via PXE → OS installs → Domain joined → Office + Chrome installed automatically.



8. Logs & Troubleshooting in SCCM OSD

When deploying Operating Systems (OSD), logs help **monitor progress** and **fix issues**.

♦ Important Client Logs (on the PC being imaged)

1. **smsts.log** → Main OSD log (captures Task Sequence progress/errors).

○ Location:

■ During WinPE: `X:\Windows\Temp\SMSTSLog\smsts.log`

■ After OS installed: `C:\Windows\CCM\Logs\smsts.log`

2. **smspxe.log** → PXE boot process (on Distribution Point server).

3. **ccmsetup.log** → Client installation logs after OS deployment.

♦ Important Server Logs

- **distmgr.log** → Content distribution to DP.
- **smsexec.log** → Task sequence execution logs.
- **wds.log** → Windows Deployment Services PXE boot log.

◇ Troubleshooting Common Issues

1. **PXE Boot Fails** → Check **smspxe.log** & DHCP Options (66, 67).
2. **Task Sequence Stops** → Review **smsts.log**.
3. **Driver Issues** → Ensure drivers are added to DP & linked to Task Sequence.
4. **Client Doesn't Install SCCM Agent** → Check **ccmsetup.log**.

◇ Example

👉 If a Windows 11 OSD fails at 50% → Check **smsts.log** → Error shows missing driver → Fix by adding driver to SCCM & re-deploy.



Module 9: Endpoint Protection in SCCM

1. Overview

- **Endpoint Protection (EP)** in SCCM helps **protect client devices** from malware, spyware, and security threats.
- It integrates with **Microsoft Defender Antivirus** (Windows 10/11).
- Admins can manage antivirus policies, scan schedules, and monitor threat reports — all from the **SCCM Console**.

◇ Key Features of Endpoint Protection

1. **Centralized Management** → Configure and deploy AV policies from SCCM.
2. **Automatic Updates** → Keeps virus definitions (signatures) updated.
3. **Real-time Protection** → Blocks malware instantly.
4. **Reports & Alerts** → Shows infected devices, threat types, and compliance.
5. **Integration** → Works with SCCM collections (e.g., HR laptops, IT Dept PCs).

◇ Example

👉 IT enables Endpoint Protection → Deploys policy to *All Windows 11 PCs* → Defender auto-scans daily → SCCM reports show if any device is infected.

Endpoint Protection (EP) in SCCM

- Endpoint Protection (EP) in SCCM protects devices from malware, spyware, viruses.



2. Configure & Monitor Endpoint Protection Policies in SCCM

◇ 1. Configuring Policies

- Go to **SCCM Console** → **Assets and Compliance** → **Endpoint Protection** → **Antimalware Policies**.
- Create a new **Antimalware Policy**.
- Configure:
 - **Scan Schedule** (daily quick scan, weekly full scan).
 - **Exclusions** (e.g., database files, system folders).
 - **Real-Time Protection** (on/off).
 - **Cloud Protection** (optional).

- Deploy the policy to a **Device/User Collection**.

👉 Example: *HR Department Laptops* → Daily quick scan + weekly full scan policy.

◇ 2. Monitoring Policies

- Go to **Monitoring** → **Security** → **Endpoint Protection Status**.
- Track:
 - Compliance (devices with updated definitions).
 - Threats detected.
 - Devices with outdated/missing policies.

👉 Example: Admin sees 95% laptops compliant, 5% outdated.

◇ 3. Logs for Troubleshooting

- **epmgr.log** → Endpoint Protection manager log (on site server).
- **EndpointProtectionAgent.log** → On client device, shows policy application.



Module 10: Compliance Settings in SCCM

1. Overview

- **Compliance Settings** in SCCM help ensure that devices in your organization follow **security policies, configurations, and standards**.
- Used to check if devices meet requirements like:
 - Antivirus enabled
 - Firewall turned on
 - Specific registry setting applied
 - Certain software installed

◇ Key Components

1. **Configuration Items (CI)**
 - Define what to check (e.g., Registry key, File existence, App version).
2. **Configuration Baselines (CB)**
 - Group of Configuration Items.
 - Deployed to a collection of devices/users.
3. **Compliance Evaluation**
 - SCCM clients evaluate → Report compliance status (Compliant / Non-compliant).
4. **Remediation**
 - If possible, SCCM can fix non-compliant settings automatically.

◇ Example

👉 Create a Compliance Baseline → Check if **Windows Firewall = ON**.

- Compliant = Green ☒
- Non-Compliant = Red ☐ → SCCM auto-enables Firewall.



2. Create Configuration Items (CI) & Baselines (CB)

◇ 1. Configuration Items (CI)

- Define *what to check or enforce* on client devices.
- Can check:
 - Registry settings
 - File/Folder existence
 - Software version
 - Script output (PowerShell/VB)

- Security settings (Firewall, Defender, etc.)

👉 Example CI: “Check if Firewall is ON”.

◇ 2. Configuration Baselines (CB)

- A collection of CIs grouped together.
- Baselines are deployed to device/user collections.
- Used for compliance evaluation.

👉 Example CB:

- CI-1 = Firewall ON
- CI-2 = Windows Defender Enabled
- CI-3 = Windows Update Service Running

= **Security Baseline.**

◇ Workflow

1. Create CI (e.g., Firewall ON).
2. Add CI to CB (e.g., Security Baseline).
3. Deploy CB to a device collection (e.g., HR Laptops).
4. Client evaluates & reports compliance.
5. Non-compliant devices can be remediated automatically.



3. Detection & Remediation Methods in Compliance Settings

◇ 1. Detection Methods

These define *how SCCM checks compliance* on client PCs.

- Registry Detection** → Checks if registry key/value exists.
 Example: `Firewall = ON.`
- File/Folder Detection** → Checks if a file or folder exists.
 Example: `C:\Program Files\Chrome\chrome.exe.`
- WMI Query** → Checks system configuration via WMI.
 Example: Confirm BIOS version.
- Script-Based Detection** → PowerShell/VBScript checks advanced conditions.
 Example: Script checks if specific Windows service is running.

◇ 2. Remediation Methods

If a device is **non-compliant**, SCCM can fix it automatically.

- Registry Fix** → Modify registry values.
- File Replacement** → Copy/install missing files.

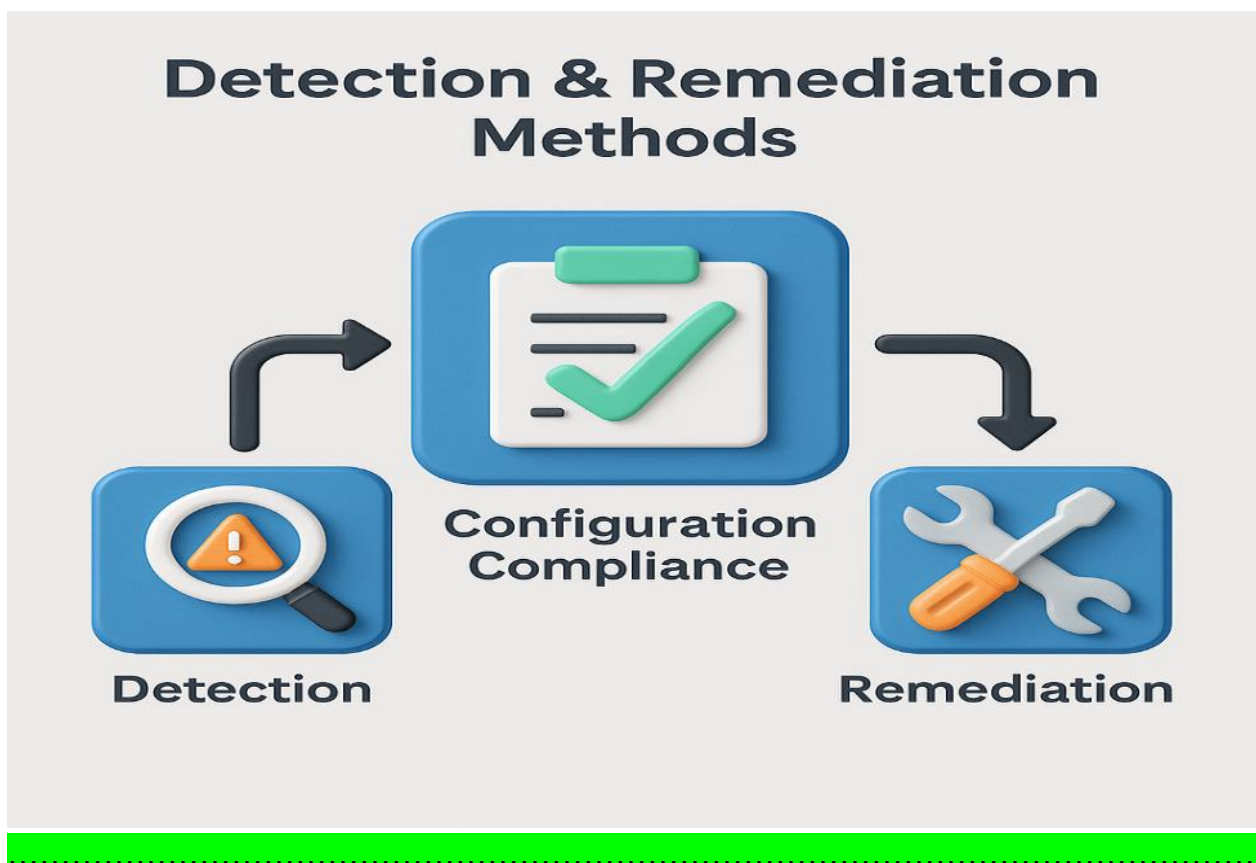
- **Service Restart/Enable** → Start or enable Windows services.
- **Script-Based Remediation** → Run script to enforce compliance.

👉 Example:

- Detection → Firewall OFF ❌
- Remediation → SCCM turns Firewall ON ✅

◇ Workflow

1. Detection → Client checks compliance (Firewall, files, registry).
2. If Non-Compliant ❌ → Remediation kicks in.
3. Report → Sent to SCCM console (Compliant/Non-compliant).



4. Viewing Compliance Results in SCCM

◇ Where to View Results

1. **SCCM Console** → **Monitoring** → **Deployments**

- Shows compliance status of deployed baselines.
- Status:  Compliant /  Non-Compliant /  Unknown.

2. Reports

- SCCM → Reporting → Compliance and Settings Management.
- Prebuilt reports show compliance % by device, collection, or baseline.

3. Client Side

- On client PC → Open **Configuration Manager Properties** → **Configurations** **tab.**
- Shows compliance evaluation result.

◇ Example

👉 Deploy a **Security Baseline** (Firewall ON, Defender ON).

- SCCM Report → 85% Compliant , 15% Non-Compliant .
- Admin drills into report → Finds which laptops have Firewall OFF.

◇ Logs for Compliance

- **CIAgent.log** → Records CI processing on client.
- **DCMAgent.log** → Baseline evaluation & remediation logs.
- **StateMessage.log** → Sends compliance status back to SCCM server.



Module 11: Mobile Device Management (MDM) in SCCM

1. Overview of Mobile Device Management (MDM) in SCCM allows IT admins to manage **smartphones, tablets, and other mobile devices**.

- It ensures that corporate security policies, apps, and configurations are applied to mobile devices — just like PCs.
- SCCM integrates with **Microsoft Intune** for cloud-based MDM.

◇ Key Capabilities of MDM

1. **Device Enrollment**
 - Register mobile devices with SCCM/Intune.
2. **Policy Deployment**

- Configure Wi-Fi, VPN, Email, and Security settings.

3. **App Management**

- Deploy mobile apps (Office, Outlook, etc.) to devices.

4. **Security Management**

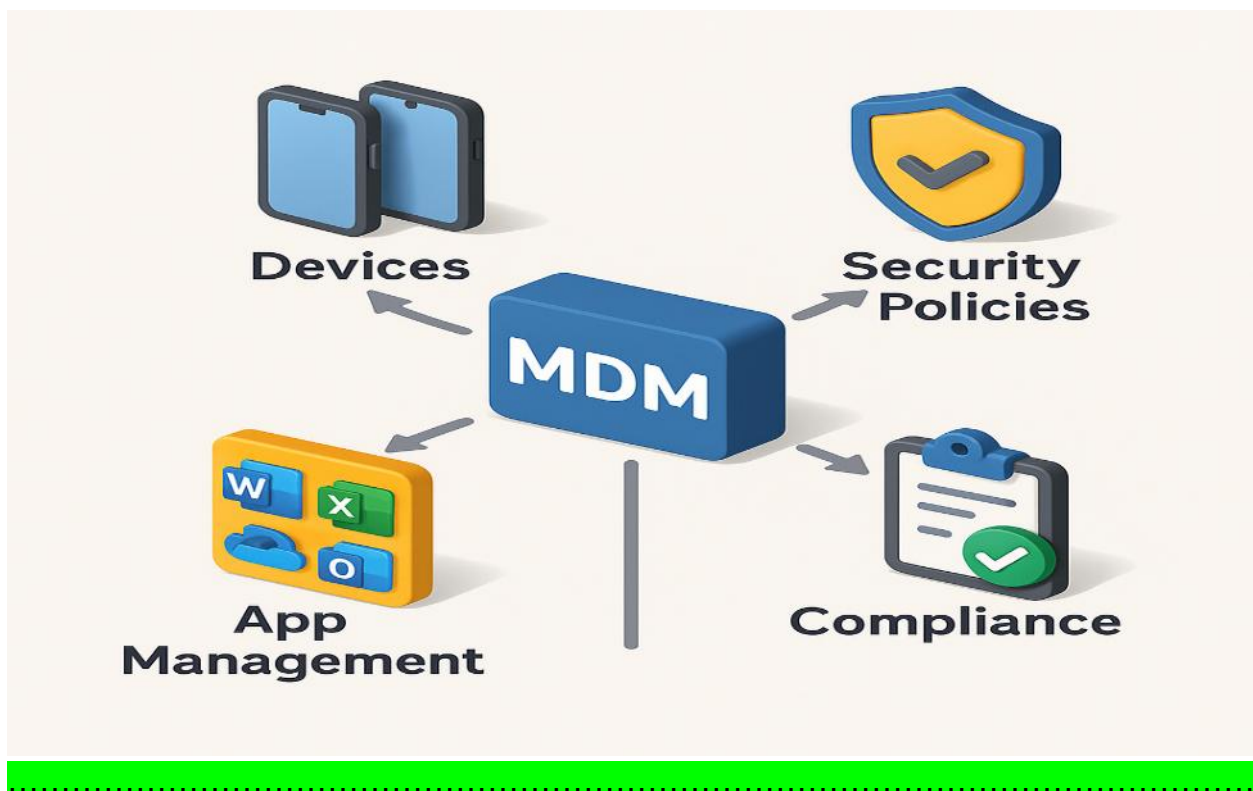
- Enforce PIN/Password policies.
- Remote wipe/lock in case of lost device.

5. **Compliance Monitoring**

- Check if device meets corporate policies (e.g., Encryption ON).

◇ **Example**

👉 HR staff enrolls their mobile phones → SCCM/Intune applies policy = PIN required, Outlook auto-installed → If phone is lost, admin can wipe it remotely.



2. Managing Mobile Devices by Using the Windows Intune Connector

◆ Explanation

- **Why Intune Connector?**
SCCM alone cannot directly manage mobile devices (Android/iOS). It needs the Intune Connector.
- **How does it work?**
 - The connector acts like a **bridge** between **SCCM (on-premises)** and **Microsoft Intune (cloud service)**.
 - When a mobile device is enrolled through **Company Portal**, it first connects to Intune (cloud).
 - Intune passes the device information back to SCCM through the connector.
- **Result:**
 - All enrolled mobile devices (phones, tablets, laptops) appear inside the **SCCM Console**.
 - Admins can now deploy apps, enforce compliance, and manage policies from SCCM.

👉 Example:

- Employee's Android phone is enrolled.
- Device info goes → Intune Cloud → Intune Connector → SCCM Console.
- IT Admin can now see and manage the device from SCCM.

◆ Flow (Step by Step)

1. SCCM (On-premises)

2. Intune Connector (Bridge)
3. Intune (Cloud)
4. Mobile Devices Enrolled
5. Devices appear in SCCM Console



3. Create Policies and Compliance Settings for Mobile Devices

◆ Explanation

- **Compliance Settings** in SCCM/Intune allow IT admins to enforce **security and usage rules** on mobile devices.
- These settings make sure all devices meet **company standards** before they can access corporate apps or data.

◆ Common Compliance Rules

1. **Minimum PIN/Password Length**

- Devices must have a strong passcode (e.g., 6-digit or longer).

2. Device Encryption Required

- All data must be encrypted to prevent unauthorized access if the device is lost/stolen.

3. Block Jailbroken/Rooted Devices

- Compromised devices (hacked for admin rights) are **not allowed** to connect.

◆ Why it's Important

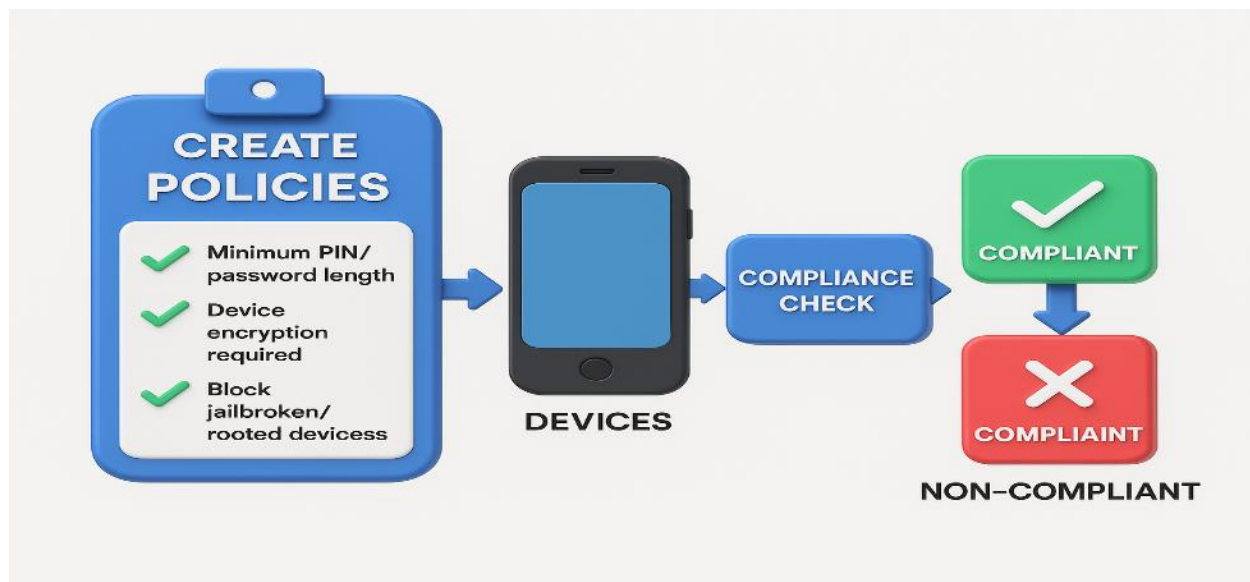
- ✓ Protects company data
- ✓ Prevents unauthorized device access
- ✓ Ensures legal and security compliance

◇ Example Scenario

- Company policy: **“All mobile devices must have a 6-digit PIN and encryption enabled.”**
- A device without encryption or with a weak password is marked **non-compliant** and access is restricted until fixed.

◆ Flow (Step by Step)

1. Admin defines compliance rules in SCCM/Intune.
2. Rules are applied to enrolled devices.
3. Devices check their settings against compliance policies.
4. If compliant → full access.
5. If non-compliant → restricted access / remediation required.



4. Enroll Mobile Devices Using Microsoft Company Portal

◆ Explanation

- **Company Portal App** is the official Microsoft app used for device enrollment into Intune/SCCM.
- Available for **iOS, Android, and Windows** platforms.

◆ Enrollment Process (Step by Step)

1. **User** **Installs** **Company** **Portal** **App**
 - From Play Store (Android), App Store (iOS), or Microsoft Store (Windows).
2. **User** **Logs** **in** **with** **Company** **Credentials**
 - Employee enters **corporate email + password** (Azure AD / Office 365 account).
3. **Device** **Enrollment** **Happens**
 - Device registers with Intune.

- Information flows back into SCCM through Intune Connector.

4. **Automatic Policies & Apps Delivery**

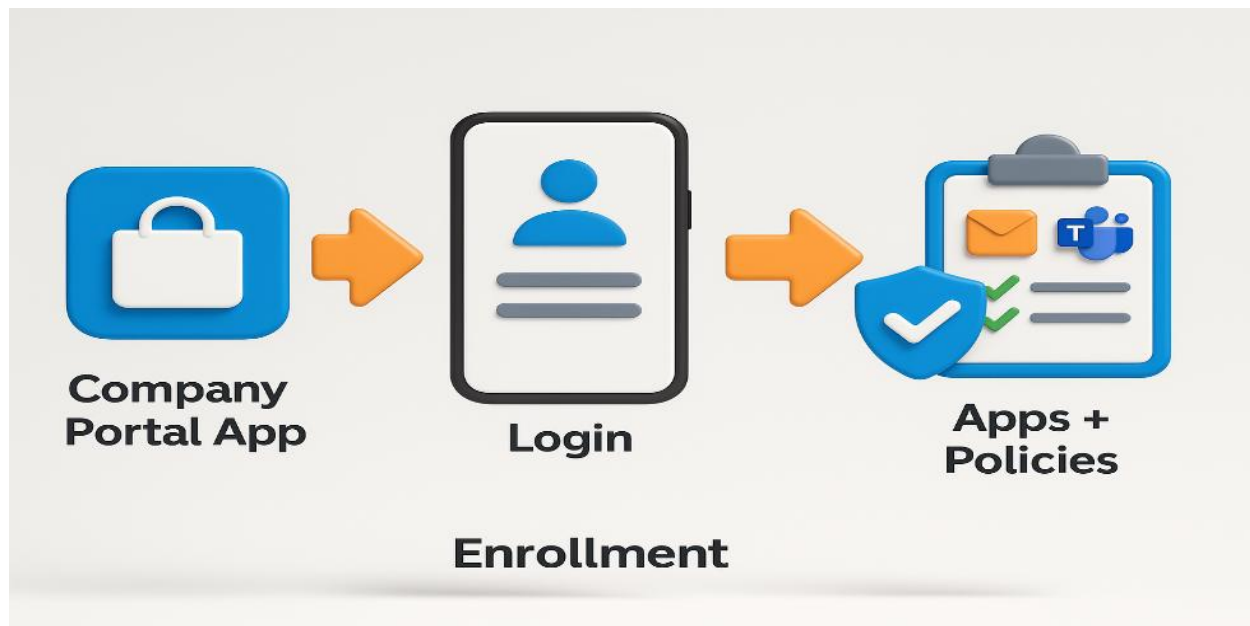
- Once enrolled, device automatically receives:
 - **Compliance policies** (PIN, encryption rules)
 - **Company apps** (Teams, Outlook, VPN, etc.)
 - **Access rules** (Wi-Fi/VPN profiles, certificates).

◇ **Example Scenario**

- A new employee joins the company.
- Installs **Company Portal App** on their phone.
- Logs in → Device enrolled automatically.
- Within minutes, their phone gets **Outlook, Teams, and security policies** applied.

☑ **Summary:**

- **Company Portal = Entry Point** for device enrollment.
- Enrollment ensures the device becomes **managed and secured**.
- Users get company apps & compliance policies automatically.



5. Deploying Applications to Mobile Devices

◆ Explanation

With SCCM integrated with **Microsoft Intune**, IT admins can deliver apps directly to mobile devices after they are **enrolled through the Company Portal**.

◆ Types of Applications You Can Deploy

1. **Microsoft Apps** → Outlook, Teams, OneDrive, Edge
2. **Store Apps** → Android Play Store / iOS App Store apps
3. **Line-of-Business (LOB) Apps** → Custom in-house apps (like HR App, Sales Tracker)

◆ Deployment Types

1. **Required** **Deployment**
 - Apps install **automatically** on user's device without asking.
 - Example: Push **Teams** to all employees.

2. Available

Deployment

- Apps appear in **Company Portal** / **Software Center**.
- Users can **choose** whether to install.
- Example: Optional apps like Adobe Reader.

◆ Example Scenario

- Admin deploys **Outlook, Teams (Required)** to all devices.
- Deploys **Power BI app (Available)** so only interested users can install.

☒ Summary of Module 11 (MDM)

- **SCCM + Intune = Full MDM support**
- **Intune Connector = Bridge for management**
- **Company Portal = Enrollment tool**
- **Compliance Policies = Enforce security**
- **App Deployment = Push Microsoft, Store, or In-house apps to devices**

